Lesson 9: Cumulative distribution functions (CDFs)

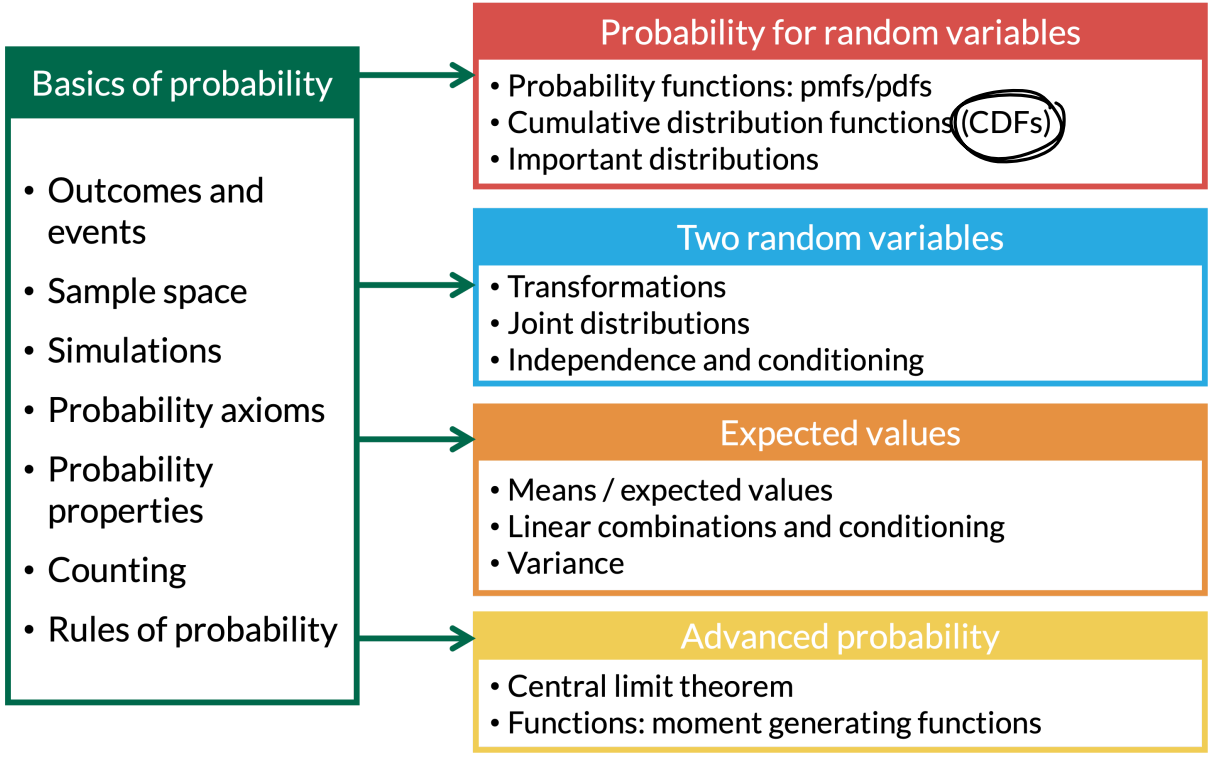
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Learning Objectives

1. Understand the definition of cumulative distribution functions (CDFs) for discrete and continuous random variables.
2. Compute CDFs for given probability mass functions (pmfs).
3. Compute CDFs for given probability density functions (pdfs) and pdfs from CDFs.
4. Simulate a ~~PDF~~ random variable with inverse CDF method.

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

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What is a cumulative distribution function?

$F_X(x) = P(X \leq x)$

Cumulative distribution function (CDF) for discrete random variable

The **cumulative distribution function (cdf)** of a **discrete RV** X with pmf $p_X(x)$, is defined for every value x by

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{\{all\ y:\ y \leq x\}} p_X(y)$$

- $F(x)$ is increasing or flat (never decreasing)
- $\min_x F(x) = 0$: the CDF starts at 0
- $\max_x F(x) = 1$: at the largest x, all the probabilities have been summed, so the total is 1
- The CDF of a discrete RV is a step function

Cumulative distribution function (CDF) for continuous random variable

The **cumulative distribution function (cdf)** of a **continuous RV** X, is the function $F_X(x)$, such that for all real values of x,

$P(X \leq 2.8) = F_X(2.8)$

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(s)ds$$

(lower bound of $f_X(x)$)

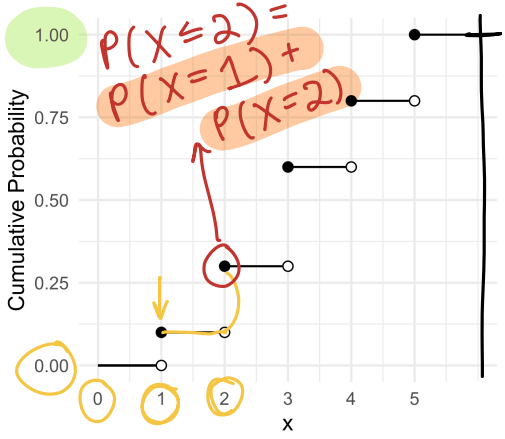
In general, $F_X(x)$ is increasing and

- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to \infty} F_X(x) = 1$
- $P(X > a) = 1 - P(X \leq a) = 1 - F_X(a)$
- $P(a \leq X \leq b) = F_X(b) - F_X(a)$

How to define CDFs for discrete and continuous RVs?

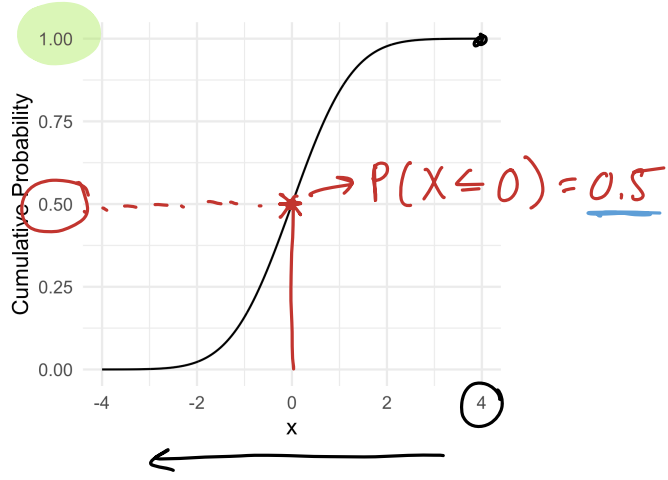
total area under curve = 1

Discrete RV X:



- pmf: $p_X(x) = P(X = x)$
- CDF: $F_X(x) = P(X \leq x) = \sum_{\{all\ y:\ y \leq x\}} p_X(y)$

Continuous RV X:



- density: $f_X(x)$
- probability: $P(a \leq X \leq b) = \int_a^b f_X(x)dx$
- CDF: $F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(s)ds$

Learning Objective

1. Understand the definition of cumulative distribution functions (CDFs) for discrete and continuous random variables.
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Example 1: Falls in Older Adults

A major public health concern is falls among older adults (age 65+). National data suggests that 25% of older adults will experience at least one fall within a given year. A community health program is tracking a random group of $n = 8$ older adults for one year. Assume the likelihood of falling is independent from person to person.

Let X be the random variable representing the number of individuals in this group who experience at least one fall.

1. Write the CDF of X.
2. Use R to calculate the cumulative probability for each possible value of X. (exact)
3. Plot the CDF of X.
4. Simulate X for 10000 groups and plot the approximated CDF.

Recall our pmf, since X is a binomial RV:

$$P(X = x) = \binom{8}{x} 0.25^x 0.75^{8-x}, x = 0, 1, 2, \ldots, 8$$

Example 1: Falls in Older Adults

1. Write the CDF of X.

Recall our pmf:

pmf → $P(X = x) = \binom{8}{x} 0.25^x 0.75^{8-x}, x = 0, 1, 2, \ldots,$

$$F_X(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{8}{k} 0.25^k 0.75^{8-k}$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 0.1 & 0 \leq x < 1 \\ 0.367 & 1 \leq x < 2 \\ 0.679 & 2 \leq x < 3 \\ 0.886 & 3 \leq x < 4 \\ 0.973 & 4 \leq x < 5 \\ \vdots & \vdots \\ 0.9997 & 7 \leq x < 8 \\ 1 & x \geq 8 \end{cases}$$

CDF @ 2: $F_X(2) = P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= \binom{8}{0} 0.25^0 0.75^8 + \binom{8}{1} 0.25^1 0.75^7 + \binom{8}{2} 0.25^2 0.75^6$

choose $(8, 2)$

$= 0.679$

Example 1: Falls in Older Adults

2. Use R to calculate the cumulative probability for each possible value of X. exact CDF

```r
n <- 8
p <- 0.25

falls_cdf <- tibble(
  x = 0:n,
  c_prob = pbinom(x, size = n, prob = p)
)
```

↳ for a binomial RV, $P(X \leq x)$

```r
falls_cdf
```

```
# A tibble: 9 × 2
      x c_prob
  <int>  <dbl>
1     0  0.100
2     1  0.367
3     2  0.679
4     3  0.886
5     4  0.973
6     5  0.996
7     6  1.000
8     7  1.000
9     8  1
```

$P(X \leq x)$

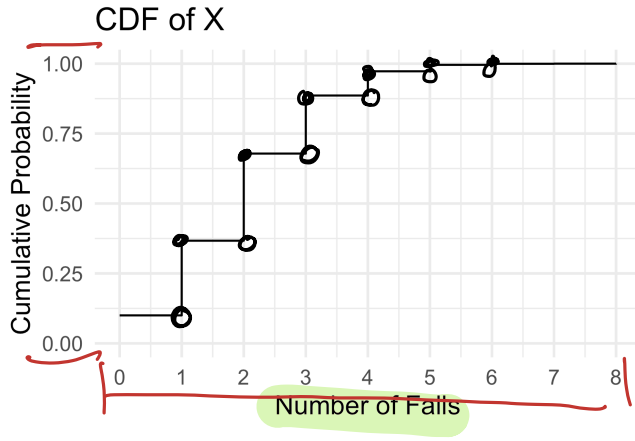
$x \to -\infty$ 0

$x \to \infty$ 1

Example 1: Falls in Older Adults

3. Plot the CDF of X. exact CDF

```
1  ggplot(falls_cdf,                                          ①
2         aes(x = x, y = c_prob)) +
3    geom_step(                                               ②
4      linewidth = 1,
5      color = "black"
6      ) +
7    scale_x_continuous(breaks = 0:n) +                       ③
8    scale_y_continuous(limits = c(0, 1)) +
9    labs(title = "CDF of X",                                 ④
10        x = "Number of Falls",
11        y = "Cumulative Probability") +
12   theme_minimal(base_size = 26)                            ⑤
```



Example 1: Falls in Older Adults

4. Simulate X for 10000 groups and plot the approximated CDF.

```
1  set.seed(4764)                                                    ①
2  reps <- 10000                                                     ②
3
4  falls_sims <- tibble(                                             ③
5    X = rbinom(n = reps, size = n, prob = p)
6    )
7
8  falls_approx <- falls_sims |>                                     ④
9    group_by(X) |>                                                  ⑤
10   summarise(prop = n() / reps) |>                                 ⑥
11   mutate(approx_cdf = cumsum(prop))                               ⑦
```

prop of reps that equal x

$P(X=x)$

- The approximate pmf and CDF:

```
1  falls_approx                                                      ①
# A tibble: 8 × 3
      X   prop approx_cdf
  <int>  <dbl>      <dbl>
1     0 0.101      0.101
2     1 0.268      0.370
3     2 0.312      0.682
4     3 0.207      0.889
5     4 0.0834     0.972
6     5 0.0243     0.997
7     6 0.0029     1.000
8     7 0.0003     1
```

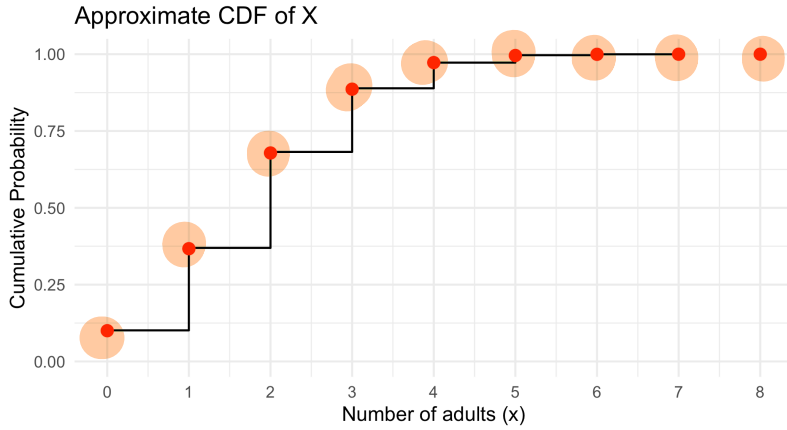
$p_X(x) = P(X=x)$

Example 1: Falls in Older Adults

4. Simulate X for 10000 groups and plot the approximated CDF.

- Red dots: the exact CDF from `pbinom()`
- Steps: the approximate CDF from the simulation
- At home: try `reps <- 100` and see how it affects the approximate CDF

▶ Plot the approximate CDF of X



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Let's demonstrate the CDF with an example

Example 2

Let $f_X(x) = 2$, for $2.5 \leq x \leq 3$. Find $F_X(x)$.

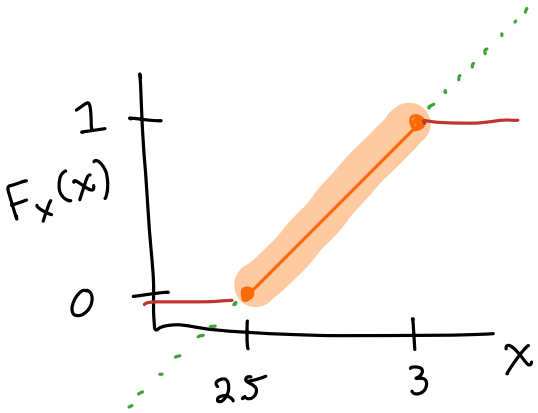
$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(s)\, ds \qquad s = x$$

$$F_X(x) = \int_{2.5}^{x} 2\, ds$$

$$= 2s \Big|_{2.5}^{s=x} = 2x - 2(2.5)$$

$$= 2x - 5 \qquad 2.5 \leq x \leq 3$$

$$F_X(x) = \begin{cases} 0 & x < 2.5 \\ 2x-5 & 2.5 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$



Derivatives of the CDF

Theorem 1

If X is a continuous random variable with pdf $f_X(x)$ and cdf $F_X(x)$, then for all real values of x at which $F'_X(x)$ exists,

$$\frac{d}{dx}F_X(x) = F'_X(x) = f_X(x)$$

$f = F'$

$\frac{dF(x)}{dx}$

- Integrate the pdf to get the CDF
- Take the derivative of the CDF to get the pdf

Finding the PDF from a CDF

Example 3

Let X be a RV with cdf

$$F_X(x) = \begin{cases} 0 & x < 2.5 \\ 2x - 5 & 2.5 \le x \le 3 \\ 1 & x > 3 \end{cases}$$

Find the pdf $f_X(x)$.

$\frac{d}{dx}(0) = 0$

$\frac{d}{dx}(2x-5) = \frac{d}{dx}(2x) - \frac{d}{dx}(5)$

$= 2 - 0 = 2$

$\frac{d}{dx}(1) = 0$

$$f_X(x) = F'(x) = \begin{cases} 0 & x < 2.5 \\ 2 & 2.5 \le x < 3 \\ 0 & x > 3 \end{cases}$$

$f_X(x) = 2$ for $2.5 \le x \le 3$

Let's go through another example (1/7)

Example 4

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

1. Show $f_X(x)$ is a pdf.
2. Find $\mathbb{P}(1 \leq X \leq 3)$.
3. Find $F_X(x)$.
4. Given $F_X(x)$, find $f_X(x)$.
5. Find $\mathbb{P}(X \geq 1 | X \leq 3)$.
6. Find the median of the distribution of X.

Let's go through another example (2/7)

Example 4.1

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

1. Show $f_X(x)$ is a pdf.

To show $f_X(x)$ is a valid pdf, we need to show two things

1. $f_X(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

If both hold, then $f_X(x)$ is a pdf

① $f_X(x) = 2e^{-2x} \geq 0$?

$x > 0$

$x = 1 : f_X(1) = 2e^{-2} = \frac{2}{e^2} > 0$

$x = 4 \quad f_X(4) = \frac{2}{e^8}$

$x \to \infty \quad f_X(x) \to 0$

$x \to 0 \quad f_X(x) = 2$

② $\int_0^{\infty} 2e^{-2x}\,dx$

$u = -2x$

$du = -2\,dx$

$-du = 2\,dx$

bounds/integrands

$u = -2(\infty) = -\infty$

$u = -2(0) = 0$

$= +\int_0^{-\infty} e^u (-du)$

$= \int_{-\infty}^{0} e^u\,du = e^u \Big|_{u=-\infty}^{u=0}$

$= e^0 - e^{-\infty} = 1 - 0 = 1$

$f_X(x)$ is a valid pdf!!

Let's go through another example (3/7) [REFERENCE]

Example 4.2

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

2. Find $\mathbb{P}(1 \leq X \leq 3)$.

Integrate the pdf from 1 to 3:

$$\mathbb{P}(1 \leq X \leq 3) = \int_1^3 2e^{-2x}\,dx$$

Use a u-substitution: let $u = -2x$. Then $du = -2\,dx$, so $2\,dx = -du$.

New bounds: when $x = 1, u = -2$; when $x = 3, u = -6$.

$$\begin{aligned}\int_1^3 2e^{-2x}\,dx &= \int_{-2}^{-6} e^u\,(-du)\\ &= -\Big[e^u\Big]_{u=-2}^{u=-6}\\ &= -\left(e^{-6} - e^{-2}\right)\\ &= e^{-2} - e^{-6} \approx 0.1329\end{aligned}$$

Check with the antiderivative: $\Big[-e^{-2x}\Big]_{x=1}^{x=3} = -e^{-6} + e^{-2}$.

Let's go through another example (4/7)

Example 4.3

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

3. Find $F_X(x)$.

Let's go through another example (5/7)

Example 4.4

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

4. Given $F_X(x)$, find $f_X(x)$.

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-2x} & x > 0 \end{cases}$$

The pdf is the derivative of the CDF: $f_X(x) = \dfrac{d}{dx} F_X(x)$.

For $x > 0$, use the chain rule on e^{-2x}:

$$\begin{aligned} f_X(x) &= \frac{d}{dx}\left(1 - e^{-2x}\right) \\ &= 0 - (-2)\, e^{-2x} \\ &= 2e^{-2x} \end{aligned}$$

For $x \leq 0$, $F_X(x) = 0$, so its derivative is 0.

$$f_X(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

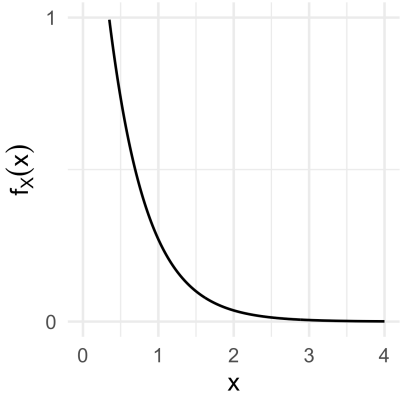
This matches the pdf we started with.

Let's go through another example (6/7)

Example 4.5

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

5. Find $\mathbb{P}(X \geq 1 | X \leq 3)$.

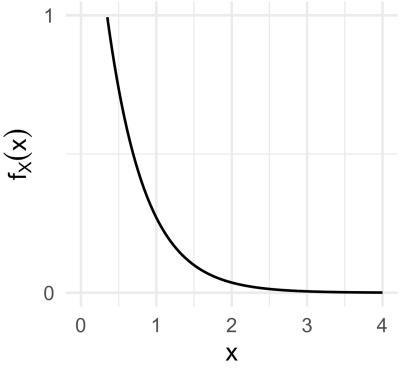


Let's go through another example (7/7)

Example 4.6

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$.

6. Find the median of the distribution of X.



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4. Simulate a CDF with inverse CDF method.

Simulating distributions R knows vs. does NOT know

- So far we simulated from distributions R already knows (`runif()`, `rbinom()`)
- What if we only have a pdf or CDF, like Example 4 with $f_X(x) = 2e^{-2x}$?

- How do we simulate X knowing $f_X(x) = 2e^{-2x}$??
  - Box method (`sample()`) won't work
  - We do not have any handy `rbinom()` functions in R for it

- We can simulate X with the inverse CDF method

Simulating from a CDF: inverse CDF method

Inverse CDF method

If U is a uniform random variable on $[0, 1]$ and F_X is a CDF, then $X = F_X^{-1}(U)$ has CDF F_X.

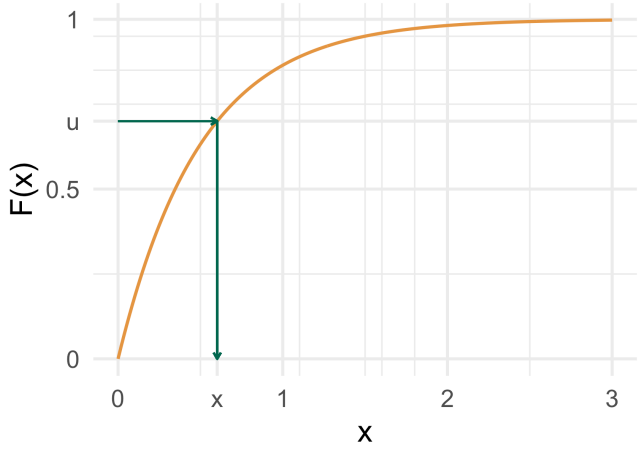
1. Find the CDF $F_X(x)$ (if starting from a pdf of X)
2. Set $u = F_X(x)$ and solve for x, to get $x = F_X^{-1}(u)$
3. Simulate U with `runif()` and plug it into F_X^{-1}

- Why does it work? For any x,

$$P(F_X^{-1}(U) \leq x) = P(U \leq F_X(x)) = F_X(x)$$

because $P(U \leq u) = u$ for a uniform U on $[0, 1]$

- Pick a random height u between 0 and 1, go across to the CDF, then down to x



Example 4 (continued)

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$, and CDF $F_X(x) = 1 - e^{-2x}$. Simulate X.

- Solve $u = 1 - e^{-2x}$ for x:

```
1 set.seed(2026)                                                    ①
2 reps <- 10000                                                     ②
3 exp_sims <- tibble(u = runif(n = reps, min = 0, max = 1)) |>      ③
4   mutate(x = -log(1 - u) / 2)                                     ④
```

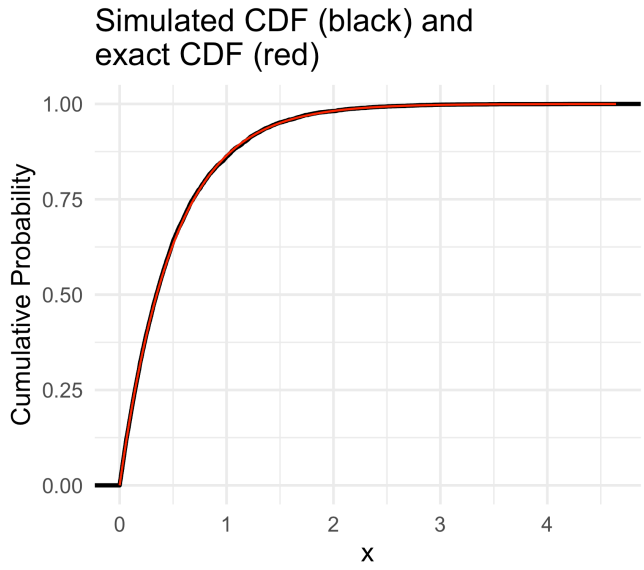
```
1 head(exp_sims)                                                    ⑤
```

```
# A tibble: 6 × 2
      u      x
  <dbl>  <dbl>
1 0.699  0.600
2 0.557  0.407
3 0.140  0.0755
4 0.286  0.168
5 0.555  0.405
6 0.0251 0.0127
```

Example 4 (continued)

Let X be a RV with pdf $f_X(x) = 2e^{-2x}$, for $x > 0$, and CDF $F_X(x) = 1 - e^{-2x}$. Simulate X.

```
1  ggplot(exp_sims, aes(x = x)) +                    ①
2    stat_ecdf(                                      ②
3      linewidth = 2, color = "black"
4      ) +
5    geom_line(                                      ③
6      aes(y = 1 - exp(-2 * x)),
7      color = "red", linewidth = 1
8      ) +
9    labs(                                           ④
10     title = "Simulated CDF (black) and \nexa
11     x = "x",
12     y = "Cumulative Probability"
13     ) +
14   theme_minimal(base_size = 26)                   ⑤
```



Check our hand-calculated answers from Example 4

```r
exp_sims |>
  summarise(
    prob_1_to_3  = mean(x >= 1 & x <= 3),
    prob_cond    = mean(x >= 1 & x <= 3) / mean(x <= 3),
    median_x     = median(x)
  )
```

```
# A tibble: 1 × 3
  prob_1_to_3 prob_cond median_x
        <dbl>     <dbl>    <dbl>
1       0.136     0.137    0.344
```

- This works for any CDF we can invert: the same three steps, a different F_X^{-1}
- The simulated values are a **transformation** of U
  - We will study transformations of random variables in the next lessons!